SkillMarket | Marketplace entities

SQLite schema verified against Django models, migrations and the local database | 27 October 2026

Relationship notation: parent (1) -- child (0..N), or (0..1). Every child FK is required. No role column exists.

auth_user

PK	id	INTEGER
	password	VARCHAR(128)
	last_login ?	DATETIME
	is_superuser	BOOL
UQ	username	VARCHAR(150)
	first_name	VARCHAR(150)
	last_name	VARCHAR(150)
	email	VARCHAR(254)
	is_staff	BOOL
	is_active	BOOL
	date_joined	DATETIME

User (1) -- Listing (0..N): seller_id

marketplace_listing

PK	id	INTEGER
FK	seller_id	INTEGER
	title	VARCHAR(200)
	description	TEXT
	category	VARCHAR(100)
	starting_price	DECIMAL
	status	VARCHAR(8)
	created_at	DATETIME
	updated_at	DATETIME

User (1) -- Offer (0..N): buyer_id

Listing (1) -- Offer (0..N): listing_id

marketplace_offer

PK	id	INTEGER
FK	listing_id	BIGINT
FK	buyer_id	INTEGER
	offered_price	DECIMAL
	message	TEXT
	status	VARCHAR(10)
	created_at	DATETIME
	updated_at	DATETIME

1 : 0..1 (offer_id UQ)

marketplace_review

PK	id	INTEGER
FK UQ	offer_id	BIGINT
FK	reviewer_id	INTEGER
FK	seller_id	INTEGER
	rating	SMALLINT UNSIGNED
	comment	TEXT
	created_at	DATETIME

User (1) -- Review (0..N): reviewer_id

User (1) -- Review (0..N): seller_id

Built-in auth.User

Buyer / Seller membership: see page 2.

All marketplace FK fields use Django PROTECT.
Database FK actions: NO ACTION (SQLite).

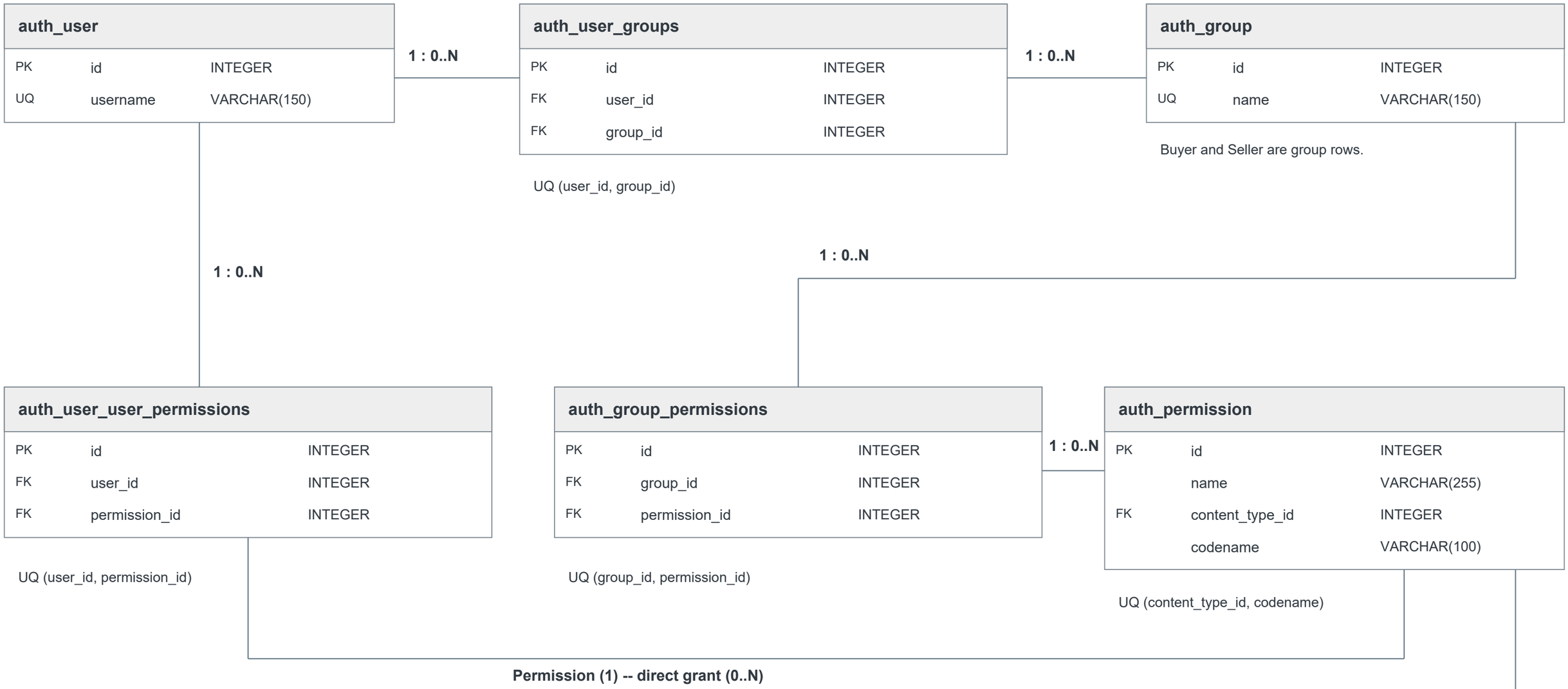
Types are actual SQLite declarations.
Domain IDs: BigAutoField -> INTEGER PK.
Domain references: BIGINT; user IDs: INTEGER.

DECIMAL fields: Django max_digits=10, decimal_places=2; SQLite does not enforce fixed decimal precision or VARCHAR lengths.

SkillMarket | Authentication and permissions

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Each connector is parent (1) -- join/child (0..N). Composite UQ constraints prevent duplicate memberships.



Role and permission implementation

0002_roles creates Buyer/Seller groups and grants model permissions.

Marketplace authorization checks group membership and record ownership.

Django Admin uses staff/model permissions. Direct user grants are supported by Django; their join table exists, but role seeding uses group grants.

SkillMarket | Constraints, indexes and verification

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Database-enforced integrity

Listing: starting_price > 0; status IN (ACTIVE, INACTIVE).

Offer: offered_price > 0; status IN (PENDING, ACCEPTED, REJECTED, COMPLETED).

Review: 1 <= rating <= 5; offer_id UNIQUE and NOT NULL; all domain foreign keys NOT NULL.

All tables shown have primary keys. auth_user.username and auth_group.name are unique.

Auth join pairs, permission (content_type_id, codename), and content type (app_label, model) are unique.

Explicit marketplace indexes (all confirmed in SQLite)

marketplace_status_a6670d_idx: Listing(status, category) - published/category discovery.

marketplace_seller__639ce1_idx: Listing(seller_id, status) - seller listings, optionally by status.

marketplace_title_7bdd5a_idx: Listing(title) - ordered/equality access; not an arbitrary substring-search accelerator.

marketplace_buyer_i_b1e5a0_idx: Offer(buyer_id, status) - buyer dashboard and status filtering.

marketplace_listing_6b4340_idx: Offer(listing_id, status) - listing offers and status filtering.

Django also creates FK indexes; Review.offer_id uses its unique index. Full inventory is in the notes.

SQLite INTEGER PRIMARY KEY uses the rowid B-tree; it does not require a separate PK index.

Application rules, defaults and delete behavior

Ownership, Buyer/Seller eligibility, valid state transitions, completed-offer reviews and cross-table author/seller consistency are checked in services.py. Foreign keys alone do not enforce these business rules.

Django defaults: Listing.status=ACTIVE; Offer.status=PENDING. Timestamps use auto_now_add / auto_now.

These are ORM behavior, not database DEFAULT clauses or automatic database update triggers.

Domain relationships: Django PROTECT. Auth/content-type relationships shown: Django CASCADE.

SQLite reports NO ACTION FK actions; Django implements PROTECT/CASCADE in its deletion collector.

Verification and known gaps

All migrations applied, including marketplace.0001_initial and 0002_roles; no pending model migrations.

Live column types, nullability, FKs, checks, unique constraints and indexes inspected; PRAGMA foreign_key_check: 0 violations.

SQLite is active locally. PostgreSQL is supported by settings but Supabase is not configured.

Category membership, text lengths, decimal precision and exact one-role-per-user are not database constraints.

Review author/seller consistency and completed status have no cross-table database check; direct writes can bypass services.

No application code, migrations or database schema changed. Internal session/admin-log tables are omitted from the main ERD.